

Fiber functors for finite symmetric tensor categories in positive characteristic

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Abstract

We prove the finite case of the Benson–Etingof–Ostrik conjecture: every finite symmetric tensor category over an algebraically closed field k of characteristic $p > 0$ admits a k -linear exact faithful strong symmetric monoidal functor to a finite higher Verlinde category $\mathrm{Ver}_{p^n}(k)$. The result includes characteristic two, and the level n may depend on the category.

Contents

1	Introduction	2
2	Finite tensor categories and products	4
3	Restricted tilting modules	5
4	From restricted tiltings to a higher Frobenius functor	10
5	A first nonprojective tensor-power module	12
6	Selecting an embedding by rank support	14
7	Removing the original category from the target	17
8	Classification, uniqueness, and reconstruction	22

1 Introduction

Let k be an algebraically closed field of characteristic $p > 0$. A *finite symmetric tensor category over k* is an essentially small k -linear abelian rigid symmetric monoidal category with biexact tensor product, finite-dimensional morphism spaces, finite-length objects, finitely many isomorphism classes of simple objects, enough projectives, and $\text{End}(\mathbf{1}) = k$. A *fiber functor* between such categories is a k -linear exact faithful strong symmetric monoidal functor. The higher Verlinde categories of Benson–Etingof–Ostrik form a tower

$$\text{Ver}_p(k) \subset \text{Ver}_{p^2}(k) \subset \text{Ver}_{p^3}(k) \subset \cdots$$

of finite symmetric tensor categories [2, 7]. The first level is the semisimplification of $\text{Rep}_k(\mathbb{Z}/p)$: one quotients by morphisms $f : X \rightarrow Y$ satisfying $\text{tr}(gf) = 0$ for all $g : Y \rightarrow X$. Its simple objects are represented by Jordan blocks of lengths $1, \dots, p-1$ [18, Section 2.3]. Thus Ver_2 is the category of vector spaces and Ver_3 is the category of supervector spaces with its sign symmetry. The higher levels need not be semisimple. They are finite abelian envelopes of quotients of the category of tilting SL_2 -modules (modules with both Weyl and dual Weyl filtrations), where the tensor ideal generated by the Steinberg module of highest weight $p^n - 1$ is killed at level n [2, Sections 3.1 and 4.1, Theorem 4.2]. Our main result is the following.

Theorem 1.1. *For every prime p , every algebraically closed field k of characteristic p , and every finite symmetric tensor category $\mathcal{C}$ over k , there are an integer $n \geq 1$ and a fiber functor*

$$\mathcal{C} \longrightarrow \text{Ver}_{p^n}(k).$$

This proves the finite case of the Benson–Etingof–Ostrik conjecture [2, Conjecture 1.4], including characteristic two. The full conjecture concerns symmetric tensor categories of *moderate growth*: for each object, the lengths of its tensor powers grow at most exponentially. Its target is the union of the higher Verlinde tower. Here both the projective generator and the finite set of simple classes are essential. The theorem places the image in a single finite level, whose height may depend on $\mathcal{C}$; it makes no assertion about arbitrary nonfinite categories of moderate growth.

A finite symmetric tensor category is *incompressible* if every k -linear exact symmetric tensor functor out of it is fully faithful. Theorem 1.1 and the higher Verlinde subcategory classification of Benson–Etingof–Ostrik [2] classify these finite categories, giving the finite case of Coulembier–Etingof–Ostrik’s Conjecture B [9]. In characteristic two, their subterminality theorem further gives uniqueness up to tensor isomorphism of the functor into the tower union. Their generalized Tannaka theorem also realizes every finite category as representations of an affine group scheme internal to a chosen finite level, with the canonical fundamental-group compatibility. Section 8 gives the precise statements and deductions.

Background and contribution

In characteristic zero, Deligne’s theorem makes supervector spaces a common target for fiber functors from symmetric tensor categories of moderate growth [12]. In positive characteristic, Ostrik proved that symmetric fusion categories admit fiber functors to Ver_p [19, Theorem 1.5]. Etingof–Ostrik constructed the ordinary Frobenius functor for nonsemisimple categories; Coulembier independently constructed its underlying additive functor as the external Frobenius twist [6, Section 4.3]. Etingof–Ostrik established its symmetric monoidal structure and characterized the finite categories on which it is exact by the existence of a fiber functor to Ver_p [18, Proposition 5.1

and Theorem 8.1]. Coulembier–Etingof–Ostrik extended that characterization to categories of moderate growth [8].

The higher levels are needed because the ordinary Frobenius functor need not be exact. The characteristic-two examples of Benson–Etingof [1] led to the all-prime tower of Benson–Etingof–Ostrik [2], constructed independently through abelian envelopes by Coulembier [7]. Coulembier–Flake developed higher Frobenius functors from elementary abelian group representations [11]. Their construction becomes symmetric monoidal once a specific restricted-tilting conjecture is established. Besides the rank-one case, the conjecture was already known for $p = r = 2$ [11, Remark 5.4.4(1) and Section 6]. Their subsequent work proved that assertion for substantial classes of modules, including modules of Loewy length at most two and Carlson modules, and for the subcategory they generate under tensor products, duals, syzygy shifts, finite sums, and summands [10, Theorem 2]. For rank two they also established the assertion for every cyclic summand of the relevant Steinberg tensor product [10, Theorem 3].

Our independent representation-theoretic contribution is the full restricted-tilting conjecture, proved in Theorem 3.1. For $r \geq 1$, embed $E_r = (\mathbb{Z}/p)^r$ in the upper unipotent subgroup of $SL_2(k)$, and write R for restriction along this embedding. Let St_{r-1} be the simple SL_2 -module of highest weight $p^{r-1} - 1$. For every finite-dimensional E_r -module D , we show that

$$R(\text{St}_{r-1}) \otimes D \cong R(T)$$

for a tilting SL_2 -module T . This is [11, Conjecture 5.4.2(1)], also formulated as [10, Conjecture 1]. The proof lifts D to the additive group, extends the associated equivariant vector bundle from the punctured plane across the origin, and uses a formal comparison of fibers after tensoring by the Steinberg module. The bounded-weight tilting input is the established [2, Lemma 3.3], in the extension-closed form also used in [10, proof of Lemma 3.1.6]. The geometric comparison is the additional step that treats arbitrary D .

Proof strategy

The restricted-tilting theorem supplies the hypothesis for the Coulembier–Flake construction. For each embedding of E_r as above, Section 4 therefore produces an additive strong symmetric monoidal functor

$$\Phi : \mathcal{C} \longrightarrow \mathcal{C} \boxtimes \text{Ver}_{p^r}(k).$$

Here $\boxtimes$ denotes the Deligne tensor product of finite abelian categories. The functor is initially semilinear: $\Phi(af) = a^{p^r} \Phi(f)$ for $a \in k$ and a morphism f . Its exactness depends on the embedding.

To select an exact functor, choose a projective generator P of $\mathcal{C}$. The regular permutation action of E_r on p^r letters makes $\text{Hom}(P, P^{\otimes p^r})$ an E_r -module. Section 5 proves that this module is nonprojective for some r , using projectivity detection and the symmetrizer criterion of [8, Lemma 4.10]. At the least such r , rank support selects an embedding for which Φ is exact (Theorem 6.5 in Section 6).

It remains to remove the $\mathcal{C}$ factor. Section 7 compares the Frobenius–Perron dimensions on that side of Φ : if S is simple and $A \boxtimes B$ is a simple constituent of $\Phi(S)$, then the dimension of A is at most that of S . Equality forces S into the Frobenius-exact subcategory of $\mathcal{C}$ and makes A the only possible first component. In the finite graph with arrows $S \rightarrow A$, every cycle therefore lies in that subcategory and cannot be left. After finitely many iterations, all first components lie there, so the image lies in its Serre closure on the first side. This adapts the finite-simple descent of [18, Section 10] to the exact higher functor. The Serre-closure results of that paper provide a fiber

functor to Ver_p in odd characteristic and to $\text{Ver}_4^+ \subset \text{Ver}_4$ in characteristic two. Combining the resulting factors in one tower level and correcting the scalar semilinearity completes Theorem 1.1.

2 Finite tensor categories and products

Throughout, k is algebraically closed of characteristic $p > 0$. Unless explicitly stated otherwise, group representations and rational algebraic-group modules are finite dimensional over k . A tensor functor is strong monoidal; when called symmetric it also preserves the given symmetries. We first relate the hypotheses of Theorem 1.1 to the usual finite-tensor-category conventions.

2.1 The finite-category conventions

Lemma 2.1. *Under the conventions of Theorem 1.1, the unit of $\mathcal{C}$ is simple and $\mathcal{C}$ has a projective generator P . The functor $\omega = \text{Hom}_{\mathcal{C}}(P, -)$ is exact and faithful, and $\mathcal{C}$ is equivalent as a k -linear abelian category to the category of finite-dimensional modules over a finite-dimensional k -algebra.*

Proof. We recall the unit argument of Deligne–Milne [13, Proposition 1.17]. Let $i : U \hookrightarrow \mathbf{1}$ and $V = \text{coker } i$. The epimorphism $V \otimes U \leftarrow U$ induced by $\mathbf{1} \twoheadrightarrow V$ has zero composite with the monomorphism $V \otimes i : V \otimes U \rightarrow V$. Thus $V \otimes U = 0$. Put $W = \ker(i^* : \mathbf{1} \rightarrow U^*)$. For a subobject $j : T \hookrightarrow X$, adjunction and exactness give

$$T \otimes U = 0 \iff (i^* \otimes \text{id}_X)j : T \longrightarrow U^* \otimes X \text{ is zero.}$$

Consequently $W \otimes X$ is the largest subobject of X annihilated by tensoring with U . Taking $X = \mathbf{1}, V$ gives $W \otimes U = 0$ and $W \otimes V \simeq V$. Tensoring $0 \rightarrow U \rightarrow \mathbf{1} \rightarrow V \rightarrow 0$ with W now shows that the natural map $W \rightarrow V$ is an isomorphism. Hence $\mathbf{1} = U \oplus W$. Since $\text{End}(\mathbf{1}) = k$ has no nontrivial idempotents, $\mathbf{1}$ is simple.

Choose representatives $S_1, \dots, S_t$ of the simple objects and projective objects Q_i with epimorphisms $Q_i \twoheadrightarrow S_i$. Set $P = \bigoplus_i Q_i$. Induction on length, using projectivity to lift an epimorphism onto a simple quotient, shows that every object is a quotient of a finite direct sum of copies of P . Thus P is a projective generator. Evaluation by P is exact; it is faithful because a nonzero image of a morphism receives a nonzero map from P , which lifts across the epimorphism onto that image. Writing $A = \text{End}(P)^{\text{op}}$, finite presentations by copies of P identify $\mathcal{C}$ with A -modules of finite dimension. Indeed, evaluation is fully faithful on such presentations, and every finite A -module has a presentation by finite free A -modules. The algebra A is finite-dimensional by the Hom-space hypothesis. $\square$

We fix such a P . In particular there is an epimorphism $P \twoheadrightarrow \mathbf{1}$: the unit is one of the S_i . The finite-algebra description also supplies projective covers of all simple objects whenever they are needed below.

2.2 Deligne products

For finite k -linear abelian categories $\mathcal{A}$ and $\mathcal{B}$, their Deligne product $\mathcal{A} \boxtimes \mathcal{B}$ represents k -bilinear functors that are right exact in each variable. If $\mathcal{A} \simeq A\text{-mod}$ and $\mathcal{B} \simeq B\text{-mod}$, then $\mathcal{A} \boxtimes \mathcal{B} \simeq (A \otimes_k B)\text{-mod}$. Since k is algebraically closed, its simple objects are the exterior products of simple objects of the two categories. Deligne products of finite tensor categories carry the componentwise tensor structure.

Lemma 2.2. *Let $\mathcal{A}_1, \dots, \mathcal{A}_s, \mathcal{Z}$ be finite symmetric tensor categories. Exact k -linear symmetric tensor functors $F_i : \mathcal{A}_i \rightarrow \mathcal{Z}$ induce an exact k -linear symmetric tensor functor*

$$H : \mathcal{A}_1 \boxtimes \cdots \boxtimes \mathcal{A}_s \rightarrow \mathcal{Z}, \quad H(X_1 \boxtimes \cdots \boxtimes X_s) = \bigotimes_i F_i(X_i).$$

Deligne products of exact tensor functors are exact as well.

Proof. The displayed multilinear functor extends to a right-exact functor by the universal property. Symmetry in $\mathcal{Z}$ supplies the tensor constraint by reordering the factors. The constraint, its inverse, and the coherence identities extend from exterior products by that same property. The resulting strong monoidal functor preserves duals. Dualizing a short exact sequence, applying right exactness, and dualizing back proves left exactness. The identical argument, with the componentwise tensor structure, proves the last assertion. $\square$

3 Restricted tilting modules

Let $G = \mathrm{SL}_{2,k}$, let $V = k^2$ be its standard module, and set

$$u(t) = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}.$$

Let $U \simeq \mathbb{G}_a$ be the upper unitriangular subgroup, parameterized by these matrices. Write $L(h)$ for the simple rational G -module of highest weight $h \geq 0$, and $\mathrm{St}_m = L(p^m - 1)$ for the m th Steinberg module; in particular $\mathrm{St}_0 = k$. The costandard and Weyl modules are $\nabla(a) = \mathrm{Sym}^a(V^*)$ and $\Delta(a) = \nabla(a)^*$. Here $V \simeq V^*$ by the invariant determinant pairing. A module is *tilting* if it has both a filtration by costandard modules and a filtration by Weyl modules. Denote by $T(a)$ the indecomposable tilting module of highest weight a .

Theorem 3.1. *Let $r \geq 1$, let $E_r = (\mathbb{Z}/p)^r$, and let $\iota : E_r \hookrightarrow U(k)$ be any embedding. Write R_ι for restriction along ι . For every finite-dimensional kE_r -module D , there is a tilting G -module T such that*

$$R_\iota(\mathrm{St}_{r-1}) \otimes D \simeq R_\iota(T).$$

This proves the restricted-Steinberg assertion of [10, Conjecture 1], for every embedding and every prime. The proof has three steps. We lift D to a vector bundle on the plane with bounded highest weights at the origin. Tensoring with St_{r-1} makes that origin fibre tilting. Finally, an equivariant formal comparison identifies its restriction to U with the fibre on the punctured plane. Write $R = R_\iota$ within this section.

3.1 A vector bundle with bounded origin weights

For a finite-dimensional rational U -module D , its *degree* is the largest degree of an entry of its action matrix $\rho_D(u(t))$. This number is independent of the chosen basis. The zero module will always satisfy every degree bound below.

Keep the embedding $\iota : E_r \hookrightarrow U(k)$ fixed, and put $q = p^r$. We will replace an arbitrary kE_r -module by a G -module at the origin of the standard plane. The two modules need not be isomorphic on restriction; their comparison will come after tensoring with a Steinberg module.

Proposition 3.2. *For every finite-dimensional kE_r -module D , there is a G -equivariant vector bundle $\mathcal{M}$ on $\mathbb{A}_k^2$ such that the stabilizer action on the fibre at $e_1 = (1, 0)$ restricts to D along ι . Every simple composition factor of the rational G -module $M_0 = \mathcal{M}|_0$ has highest weight less than q .*

Proof. We first lift D to a rational U -module of degree less than q . Let $H \subset k$ be the image of E_r under its additive embedding. The vector space

$$C_q = k[t]_{<q} = \text{span}_k\{1, t, \dots, t^{q-1}\}$$

is a subcoalgebra of $k[t]$, with comultiplication $\Delta f(t, s) = f(t+s)$ and counit $f \mapsto f(0)$. Evaluation induces a coalgebra map

$$\text{ev}_H : C_q \longrightarrow k^H = \mathcal{O}(E_r).$$

It is bijective: a polynomial of degree less than q vanishing at the q distinct elements of H is zero, and both spaces have dimension q . Transporting the $\mathcal{O}(E_r)$ -coaction of D across this isomorphism and then including $C_q \subset k[t]$ gives a rational U -module $\widehat{D}$ of degree less than q . Its restriction is D . This is a lift of individual representations; we do not require the lifting operation to preserve tensor products.

The orbit map $G \rightarrow \mathbb{A}^2 \setminus \{0\}$, $g \mapsto ge_1$, is a right U -torsor. Indeed its fibres are right U -cosets, and on the opens $x \neq 0$ and $y \neq 0$ it has the respective sections

$$\begin{pmatrix} x & 0 \\ y & x^{-1} \end{pmatrix}, \quad \begin{pmatrix} x & -y^{-1} \\ y & 0 \end{pmatrix}.$$

Form the associated vector bundle

$$\mathcal{E} = G \times^U \widehat{D}, \quad (g, d) \sim (gu, u^{-1}d),$$

on this punctured plane, with G acting by left multiplication. The fibre identification $d \mapsto [1, d]$ at e_1 gives precisely the prescribed U -action on $\widehat{D}$.

We next extend the bundle and its equivariance across the origin. A coherent extension $\mathcal{F}$ of $\mathcal{E}$ exists on the noetherian affine plane. Its double dual $\mathcal{M} = \mathcal{F}^{**}$ is coherent and reflexive, and still restricts to $\mathcal{E}$. On a regular surface a reflexive coherent sheaf is locally free: its stalk has depth two at a closed point, so the Auslander–Buchsbaum formula gives projective dimension zero; the stalks of dimensions zero and one are free as well. These standard extension statements can be found in [21, Tags 0G41, 0AY4, and 0B3N].

Write $a : G \times \mathbb{A}^2 \rightarrow \mathbb{A}^2$ for the action and π_2 for the projection. Over the punctured plane the bundle action is an isomorphism $\pi_2^* \mathcal{M} \rightarrow a^* \mathcal{M}$. Both sheaves are vector bundles on the regular variety $G \times \mathbb{A}^2$. Their homomorphism sheaf is locally free, and its sections extend uniquely across $G \times \{0\}$, which has codimension two. This is the usual extension property for reflexive sheaves [21, Tag 0EBJ]. Extend the isomorphism and its inverse; their composites are the identity by uniqueness. The identity and cocycle conditions hold on the punctured plane and therefore, by the same uniqueness on $\mathbb{A}^2$ and $G \times G \times \mathbb{A}^2$, hold everywhere. Thus $\mathcal{M}$ is a G -equivariant vector bundle.

It remains to bound the highest weights of M_0 . The line $\ell = ke_1$ is fixed pointwise by U . For $a \in k^\times$ put $h_a = \text{diag}(a, a^{-1})$. The bundle action of h_a identifies the fibres at e_1 and ae_1 , and

$$h_a^{-1}u(t)h_a = u(a^{-2}t).$$

Consequently the U -action on every nonzero fibre of $\mathcal{M}|_\ell$ has degree less than q .

To pass this bound to the origin, let $A = \Gamma(\ell, \mathcal{E}nd(\mathcal{M}|_\ell))$. Since U acts trivially on the base ℓ , its bundle action is an element of $A \otimes_k k[t]$, say $\sum_{j=0}^N b_j t^j$. For $j \geq q$, the section b_j vanishes on every nonzero fibre. The field k is infinite and $\mathcal{E}nd(\mathcal{M}|_\ell)$ is locally free, so $b_j = 0$. Specialization at zero therefore gives an action on M_0 of degree less than q . This bound passes to submodules and quotients by writing the action in a basis adapted to a submodule.

Finally, if $L(h)$ is the simple G -module of highest weight h , its U -action has degree exactly h . To see this, write $h = \sum_i h_i p^i$ with $0 \leq h_i < p$. Steinberg's tensor product theorem, in the form recalled in [15, Lemma 1.1 and the proof of Theorem 2.1], gives

$$L(h) \cong \bigotimes_i (\mathrm{Sym}^{h_i}(k^2))^{(i)}.$$

Here (i) denotes Frobenius twist. The action on the i th factor has degree $h_i p^i$. On the tensor product, the coefficient from $\bigotimes_i e_2^{h_i}$ to $\bigotimes_i e_1^{h_i}$ is t^h , so the upper bound h is attained. Hence every simple factor $L(h)$ of M_0 satisfies $h < q$, as required. The argument applies unchanged when $p = 2$. $\square$

3.2 The origin fibre after tensoring with a Steinberg module

We next convert the highest-weight bound on the origin fibre into a tilting module. We use the theory of tilting modules developed by Donkin [14]. Tilting modules are closed under extensions and tensor products. For $0 \leq a < p$ one has $L(a) = T(a)$, and Donkin's formula is

$$T(a) \otimes T(b)^{(1)} \simeq T(a + pb) \quad (p-1 \leq a \leq 2p-2, \quad b \geq 0), \quad (3.1)$$

where (1) means pullback along the Frobenius morphism of G . The identity $L(a) = T(a)$ in the stated range and Formula (3.1) hold in every positive characteristic, including 2; see [2, Proposition 3.1].

The following is the extension-closed form of [2, Lemma 3.3], also used in the proof of [10, Lemma 3.1.6]. We recall its short proof to make the highest-weight bound explicit.

Lemma 3.3. *Let k be algebraically closed of characteristic $p > 0$, let $m \geq 0$, and let M be a finite-dimensional rational $SL_2(k)$ -module. If every simple composition factor of M is $L(h)$ with $0 \leq h < p^{m+1}$, then $\mathrm{St}_m \otimes M$ is tilting. Here $\mathrm{St}_0 = k$.*

Proof. Tensoring a composition series of M by St_m and splicing Weyl and costandard filtrations reduces the assertion to $M = L(h)$. We induct on m . When $m = 0$, the inequality $h < p$ gives $\mathrm{St}_0 \otimes L(h) = T(h)$. For $m > 0$, write $h = d + ph'$ with $0 \leq d < p$ and $h' < p^m$. Steinberg's tensor product theorem gives

$$\mathrm{St}_m \otimes L(h) \simeq L(d) \otimes \mathrm{St}_1 \otimes (\mathrm{St}_{m-1} \otimes L(h'))^{(1)}.$$

By induction, the module in parentheses is a direct sum of tilting modules $T(b)$. Formula (3.1), applied with $a = p-1$, shows that tensoring its Frobenius twist with St_1 is tilting. Tensoring once more with $L(d) = T(d)$ preserves that property. $\square$

Apply Lemma 3.3 to the origin fibre M_0 with $m = r-1$. Its established bound $h < p^r$ shows that

$$T := \mathrm{St}_{r-1} \otimes M_0$$

is tilting. This includes $r = 1$, when the Steinberg factor is trivial. It remains to compare this origin fibre with the given module on the punctured plane.

3.3 Comparing the origin and the unipotent fibre

The bundle $\mathcal{N} = \mathrm{St}_{r-1} \otimes_k \mathcal{M}$ has the tilting origin fibre T constructed above. We now recover its representation on the punctured orbit from that fibre. The following lemma applies to any equivariant bundle with tilting origin fibre: we first trivialize it equivariantly on successive infinitesimal neighbourhoods of 0, then compare the generic point of the U -fixed line with e_1 .

Lemma 3.4 (Comparison with a tilting origin fibre). *Let k be an algebraically closed field of arbitrary positive characteristic. Let $\mathcal{N}$ be a G -equivariant vector bundle on V , and suppose that its origin fibre $T = \mathcal{N}_0$ is a tilting G -module. Then*

$$\mathcal{N}_{e_1} \cong T|_U$$

as rational U -modules over k .

Proof. We first construct compatible equivariant trivializations on every infinitesimal neighbourhood of 0. Let $A = k[x, y]$, let $\mathfrak{m} = (x, y)$, and let $N = \Gamma(V, \mathcal{N})$, a possibly infinite-dimensional rational G -module. Its quotients $N/\mathfrak{m}^j N$ are finite dimensional. Local freeness gives canonical G -isomorphisms

$$\mathfrak{m}^j N / \mathfrak{m}^{j+1} N \cong (\mathfrak{m}^j / \mathfrak{m}^{j+1}) \otimes_k T \cong \mathrm{Sym}^j(V^*) \otimes_k T \quad (j \geq 0). \quad (3.2)$$

Recall that a good filtration has costandard successive quotients, and a Weyl filtration has standard successive quotients. For $G = \mathrm{SL}_2$, the costandard module of highest weight a is $\nabla(a) = \mathrm{Sym}^a(V^*)$. Tensor products of two such modules have good filtrations. Indeed, for $a, b \geq 1$, multiplication of polynomials gives the exact sequence

$$0 \longrightarrow \mathrm{Sym}^{a-1}(V^*) \otimes \mathrm{Sym}^{b-1}(V^*) \xrightarrow{d} \mathrm{Sym}^a(V^*) \otimes \mathrm{Sym}^b(V^*) \longrightarrow \mathrm{Sym}^{a+b}(V^*) \longrightarrow 0. \quad (3.3)$$

Here $d = x_1 y_2 - y_1 x_2$ is the determinant in two copies of V^* . Multiplication by d is injective, its image lies in the kernel, and dimensions prove exactness. Induction on $\min(a, b)$, starting with a trivial factor, gives the assertion. Refining filtrations then proves the same tensor-product assertion for arbitrary modules with good filtrations. This argument is unchanged in characteristic two.

The module T has both a Weyl filtration and a good filtration. Consequently, the target of the next Ext group has a good filtration, and the standard–costandard Ext vanishing theorem of Cline–Parshall–Scott–van der Kallen gives

$$\mathrm{Ext}_G^1(T, \mathrm{Sym}^j(V^*) \otimes T) = 0 \quad (j \geq 0). \quad (3.4)$$

We use Ext in the category of rational G -modules; the vanishing follows from $\mathrm{Ext}_G^1(\Delta(a), \nabla(b)) = 0$ by induction on the two filtrations; see [20, Chapter 1, Proposition 2.6, and Chapter 4, Proposition 1.1].

Starting with the identity $s_1 : T \rightarrow N/\mathfrak{m}N$, lift successively to G -maps

$$s_j : T \longrightarrow N/\mathfrak{m}^j N \quad (j \geq 1)$$

such that s_{j+1} reduces to s_j . Pull back the short exact sequence

$$0 \longrightarrow \mathrm{Sym}^j(V^*) \otimes_k T \longrightarrow N/\mathfrak{m}^{j+1} N \longrightarrow N/\mathfrak{m}^j N \longrightarrow 0$$

along s_j . By (3.4) the resulting extension of T splits equivariantly; a splitting gives the lift s_{j+1} . Extending s_j by scalars gives an equivariant $A/\mathfrak{m}^j$ -linear map

$$(A/\mathfrak{m}^j) \otimes_k T \longrightarrow N/\mathfrak{m}^j N.$$

Both modules are free of rank $\dim_k T$ over the local Artinian ring $A/\mathfrak{m}^j$, and the map reduces to the identity on the residue field. Nakayama's Lemma therefore makes it an isomorphism. Taking the inverse limit produces an isomorphism

$$k[[x, y]] \otimes_k T \xrightarrow{\sim} \widehat{N} \quad (3.5)$$

compatible with the G -action at every finite order.

The line $y = 0$ is fixed scheme-theoretically by U . Restrict (3.5) to this line and then set $K = k((x))$. This gives a rational U_K -isomorphism

$$\varphi : T_K \xrightarrow{\sim} \mathcal{N}_{xe_1}.$$

To justify the group-scheme assertion explicitly, choose a polynomial frame for $\mathcal{N}$ on the line; this is possible because a finite projective $k[x]$ -module is free. The action is a matrix $C(x, t) \in \mathrm{GL}_{\dim T}(k[x, t])$, and the action on T is a matrix $B(t) \in \mathrm{GL}_{\dim T}(k[t])$. If $H(x)$ is the matrix of the restricted formal isomorphism, equivariance at every finite order says

$$C(x, t)H(x) = H(x)B(t) \pmod{x^j} \quad \text{for every } j \geq 1.$$

These matrices have uniformly bounded degree in t , so coefficientwise x -adic separation gives the equality in $k[[x]][t]$, and hence over $K[t]$.

We must compare the two fibres without changing the parameter of U . Put $h = \mathrm{diag}(x, x^{-1}) \in G(K)$, and let $b_h : \mathcal{N}_{e_1, K} \rightarrow \mathcal{N}_{xe_1}$ be bundle transport. Write ρ_T for the G -action on T and ρ_{e_1}, ρ_x for the two fibre actions of U . Since $hu(t)h^{-1} = u(x^2t)$, equivariance gives

$$b_h \rho_{e_1}(u(t)) = \rho_x(u(x^2t))b_h, \quad \rho_T(h)\rho_T(u(t)) = \rho_T(u(x^2t))\rho_T(h).$$

It follows that

$$f = b_h^{-1} \varphi \rho_T(h) : T_K \xrightarrow{\sim} \mathcal{N}_{e_1, K} \quad (3.6)$$

intertwines the original U -actions. Thus restriction of f to any fixed finite subgroup of $U(k)$ uses that subgroup's original embedding.

Finally, this isomorphism descends to k . In fixed k -bases, the intertwiner space

$$L = \mathrm{Hom}_U(T, \mathcal{N}_{e_1})$$

is the kernel of finitely many linear equations over k : compare the coefficients of t in the two polynomial action matrices. Hence its scalar extension to K is the intertwiner space over K . Equation (3.6) shows that the determinant polynomial on L is nonzero. Since k is infinite, this polynomial is nonzero at some point of $L(k)$, giving the required isomorphism. At no point have we asserted that the formal trivialization algebraizes on the plane. $\square$

Proof of Theorem 3.1. Given D , choose $\mathcal{M}$ as in Proposition 3.2 and put $M_0 = \mathcal{M}|_0$. Lemma 3.3 makes $\mathrm{St}_{r-1} \otimes M_0$ tilting. Apply Lemma 3.4 to $\mathcal{N} = \mathrm{St}_{r-1} \otimes_k \mathcal{M}$. Its origin fibre is the tilting module $T = \mathrm{St}_{r-1} \otimes_k M_0$. By Proposition 3.2, the fibre of $\mathcal{N}$ at e_1 restricts to $R(\mathrm{St}_{r-1}) \otimes D$ along the fixed embedding $E_r \hookrightarrow U(k)$. The comparison therefore gives

$$R(T) \cong R(\mathrm{St}_{r-1}) \otimes D,$$

as required. This comparison also applies when $r = 1$, with $\mathrm{St}_0 = k$. $\square$

4 From restricted tiltings to a higher Frobenius functor

Fix $r \geq 1$, put $q = p^r$, and choose an injective homomorphism $E = (\mathbb{Z}/p)^r \hookrightarrow U(k)$. Write R for restriction from SL_2 , and put $\mathcal{W} = \text{Ver}_{p^r}(k)$. Theorem 3.1 supplies the representation-theoretic input for the higher Frobenius construction of Coulembier and Flake. This section records that construction, explains how to evaluate it at projectives, and gives the exactness criterion we will use.

We will repeatedly use the following elementary fact. In a rigid abelian tensor category with simple unit, tensoring by a nonzero object Y is exact and faithful. Indeed, the coevaluation $1 \rightarrow Y \otimes Y^*$ is a nonzero monomorphism. Tensoring it by X shows that $X \otimes Y = 0$ forces $X = 0$. Exactness then shows that tensoring by Y detects nonzero morphisms through their images. An exact faithful functor reflects exactness as well.

4.1 The functor on group representations

Let $\mathcal{T}$ be the category of tilting SL_2 -modules, and let I_r be its tensor ideal generated by St_r . The canonical functor

$$\Sigma_r : \mathcal{T}/I_r \hookrightarrow \mathcal{W}$$

is fully faithful [2, Theorem 4.2(i)]. For every $s \geq 1$, $T(a)$ vanishes modulo I_s exactly when $a \geq p^s - 1$ [2, Section 3.1]. Stable restriction factors through a faithful functor $\mathcal{T}/I_r \rightarrow \text{Stab}(E)$ [11, Theorem 5.3.5]. Here $\text{Stab}(E)$ denotes the quotient of $\text{Rep}_k(E)$ by morphisms factoring through projective modules.

The assertion that $R(\text{St}_{r-1}) \otimes D$ is a restricted tilting module for every $D \in \text{Rep}_k(E)$ is precisely the hypothesis of [11, Conjecture 5.4.2(1), Theorem 5.4.3]. Since we have proved that assertion, the cited implication produces a k -linear strong symmetric monoidal functor

$$\Theta : \text{Rep}_k(E) \longrightarrow \mathcal{W}, \quad \Theta(R(T)) \simeq \Sigma_r(T \bmod I_r). \quad (4.1)$$

In particular, Θ is additive. It need not be exact.

Lemma 4.1. *If T is a tilting module and $R(T)$ is not projective, then $\Theta(R(T)) \neq 0$. Moreover, Θ annihilates every E -module induced from a proper subgroup of E .*

Proof. If $T \bmod I_r = 0$, stable restriction shows that $R(T)$ is projective. Thus nonprojectivity makes $T \bmod I_r$ nonzero; the fully faithful functor Σ_r and (4.1) give the first assertion.

Stable restriction also shows that $R(\text{St}_r)$ is projective. Its dimension is q , so it is the regular module: projectives over the local algebra kE are free. Equation (4.1) therefore shows that Θ kills all projectives. Set $S = R(\text{St}_{r-1})$. Its dimension p^{r-1} is strictly smaller than q , so it is nonprojective and $\Theta(S) \neq 0$.

Let $H < E$ have rank $s < r$. For $s > 0$, stable restriction at level s shows that $\text{Res}_H^E S$ is projective, because $\text{St}_{r-1} \in I_s$. The assertion is automatic for $H = 1$. For a kH -module D , the tensor identity gives

$$S \otimes \text{Ind}_H^E D \simeq \text{Ind}_H^E (\text{Res}_H^E S \otimes D),$$

which is projective. Applying Θ gives $\Theta(S) \otimes \Theta(\text{Ind}_H^E D) = 0$. Tensoring by a nonzero object of a tensor category is faithful, so $\Theta(\text{Ind}_H^E D) = 0$. $\square$

4.2 Internal evaluation

Let $\mathcal{C}$ be a finite symmetric tensor category. An object of $\mathcal{C} \boxtimes \text{Rep}_k(E)$ is an object of $\mathcal{C}$ equipped with an E -action. There is an internal extension of Θ ,

$$\Theta_{\mathcal{C}} : \mathcal{C} \boxtimes \text{Rep}_k(E) \longrightarrow \mathcal{C} \boxtimes \mathcal{W}.$$

We explain its meaning here because Θ is not assumed exact. Choose a projective generator P_0 and set $A = \text{End}_{\mathcal{C}}(P_0)^{\text{op}}$. Under $\mathcal{C} \simeq A\text{-mod}$, an E -object is an E -module with a commuting A -action. Applying Θ to its underlying E -module and to the action maps produces an A -module object in $\mathcal{W}$, hence an object of $\mathcal{C} \boxtimes \mathcal{W}$. This is the internal extension used in [11, Example 3.2.4, Proposition 3.2.6, and Section 3.3.2]; it does not require right exactness of Θ .

Lemma 4.2. *For every projective $Q \in \mathcal{C}$ and every E -object Z of $\mathcal{C}$, there is a natural isomorphism*

$$(\text{Hom}_{\mathcal{C}}(Q, -) \boxtimes \text{Id}_{\mathcal{W}})(\Theta_{\mathcal{C}}(Z)) \simeq \Theta(\text{Hom}_{\mathcal{C}}(Q, Z)). \quad (4.2)$$

Proof. The A -module corresponding to Q is a direct summand of A^m for some m . Thus $\text{Hom}_A(Q, -)$ is obtained from m copies of the underlying object by applying a split idempotent. An additive functor preserves finite direct sums and split idempotents. Applying this observation to Θ gives (4.2), including its naturality. $\square$

4.3 The tensor-power construction and exactness

Give $X^{\otimes q}$ the E -action arising from regular permutations of its tensor factors. Define

$$\Phi(X) = \Theta_{\mathcal{C}}(X^{\otimes q}) \in \mathcal{C} \boxtimes \mathcal{W}. \quad (4.3)$$

By Lemma 4.1 and [11, Theorem 5.2.1], this construction has coherent strong symmetric monoidal constraints. They come from the E -equivariant reordering $(X \otimes Y)^{\otimes q} \simeq X^{\otimes q} \otimes Y^{\otimes q}$, with the diagonal action on the right, through the internal monoidal extension of [11, Section 3.3.2]. The resulting functor is additive and satisfies

$$\Phi(af) = a^q \Phi(f) \quad (a \in k).$$

Additivity can also be seen directly from the regular action: in the expansion of $(X \oplus Y)^{\otimes q}$, every nonconstant coloring of the tensor positions has a proper stabilizer in E . The sum over its orbit is induced from that stabilizer, so is annihilated. The two constant colorings give $\Phi(X)$ and $\Phi(Y)$. The corresponding expansion on morphisms proves additivity, and scalar multiplication gives the displayed semilinearity.

Combining (4.3) and Lemma 4.2 yields

$$(\text{Hom}_{\mathcal{C}}(Q, -) \boxtimes \text{Id}_{\mathcal{W}})(\Phi(X)) \simeq \Theta(\text{Hom}_{\mathcal{C}}(Q, X^{\otimes q})). \quad (4.4)$$

Proposition 4.3. *If there are projective objects $P, Q \in \mathcal{C}$ for which*

$$\Theta(\text{Hom}_{\mathcal{C}}(Q, P^{\otimes q})) \neq 0,$$

then Φ is exact.

Proof. Equation (4.4) implies $\Phi(P) \neq 0$. Tensoring a short exact sequence by P makes it split, since tensor products of projectives with arbitrary objects are projective. The additive functor Φ preserves this split sequence. Its monoidal structure identifies the result with the image sequence tensored by $\Phi(P)$. Tensoring by the nonzero object $\Phi(P)$ is exact and faithful, and therefore reflects exactness. Thus the image sequence is exact. This argument is the nonzero-projective-image criterion of [18, Proposition 7.6 and Remark 7.7(i)], applied to the higher construction as in [11, Proposition 5.2.4]. $\square$

The next step is now concrete: we will choose r and the embedding of E so that

$$\Theta(\mathrm{Hom}_{\mathcal{C}}(P, P^{\otimes q})) \neq 0$$

for a projective generator P .

5 A first nonprojective tensor-power module

Retain the projective generator P and the exact faithful evaluation $\omega = \mathrm{Hom}_{\mathcal{C}}(P, -)$ from Lemma 2.1. We now select the rank of the elementary abelian group for the higher Frobenius construction. The essential point is that exact evaluation by a projective generator detects internal norm maps, even though that evaluation does not preserve tensor products.

5.1 Norms and independent group actions

For an object X with an action of a finite group H , denote its invariants and coinvariants by X^H and X_H . These are, respectively, the simultaneous kernel and cokernel of the maps $h - 1$. The endomorphism $\sum_{h \in H} h$ factors through a morphism

$$\nu_H(X) : X_H \longrightarrow X^H,$$

called the norm map. Exact k -linear functors preserve invariants, coinvariants, and this map.

Lemma 5.1. *Let H be a finite p -group and M a finite-dimensional kH -module. Then M is projective if and only if $\nu_H(M)$ is an isomorphism. For an arbitrary finite group G , the norm map of every projective kG -module is an isomorphism.*

Proof. For the regular module kG , both coinvariants and invariants are one-dimensional, generated by $[1]$ and $\sum_{g \in G} g$, respectively; the norm takes the first generator to the second. The assertion therefore holds for free modules and their direct summands.

For the converse, kH is a local self-injective algebra. Its unique simple module is the trivial module, and its socle is the line generated by $N_H = \sum_{h \in H} h$. The augmentation ideal is nilpotent by induction on $|H|$: choose a central element z of order p , use $(z - 1)^p = 0$, and pass to $k[H/\langle z \rangle]$. This proves locality. The nondegenerate associative pairing that takes (a, b) to the coefficient of 1 in ab identifies kH with its vector-space dual as a module, proving self-injectivity. The socle is the space of left H -invariants in the regular module, hence the stated line. Every nonzero left ideal meets the socle. Thus, if $N_H m \neq 0$, the map $kH \rightarrow M$, $a \mapsto am$, is injective and splits by self-injectivity. Removing free summands repeatedly gives $M \simeq (kH)^a \oplus M'$ with $N_H M' = 0$. If $M' \neq 0$, then $M'_H \neq 0$ by Nakayama's Lemma, so its zero norm map cannot be an isomorphism. An isomorphic norm therefore forces $M' = 0$. $\square$

Lemma 5.2. *Let finite p -groups $H_1, \dots, H_s$ act on objects $X_1, \dots, X_s$ of $\mathcal{C}$, respectively. If each $\omega(X_i)$ is projective over kH_i , then $\omega(X_1 \otimes \dots \otimes X_s)$ is projective over $k[H_1 \times \dots \times H_s]$, for the independent actions on the tensor factors. Its restriction to every subgroup of that product is projective as well.*

Proof. By Lemma 5.1, each evaluated norm $\omega(\nu_{H_i}(X_i))$ is an isomorphism. Exact faithfulness of ω implies that each internal norm is an isomorphism. Exactness of tensor products gives the natural identifications

$$(X \otimes Y)_{H \times K} \simeq X_H \otimes Y_K, \quad (X \otimes Y)^{H \times K} \simeq X^H \otimes Y^K$$

when H acts on X and K acts on Y . One obtains these identities by taking kernels or cokernels for one group at a time. Under them, $\nu_{H \times K}(X \otimes Y) = \nu_H(X) \otimes \nu_K(Y)$. Iteration proves that the product-group norm is an isomorphism. Applying ω and Lemma 5.1 proves the first claim. Restriction preserves projectivity because a finite group algebra is free over the group algebra of any subgroup. $\square$

5.2 Existence and minimality

Let $E_a = (\mathbb{Z}/p)^a$ act on p^a positions by its regular action, and hence act on $P^{\otimes p^a}$ through the symmetry of $\mathcal{C}$. Define

$$M(a) = \omega(P^{\otimes p^a}) \quad (a \geq 1).$$

Proposition 5.3. *There is a least integer $r \geq 1$ for which $M(r)$ is not projective over kE_r . For this r , the restriction of $M(r)$ to every proper subgroup of E_r is projective.*

Proof. Suppose first that every $M(a)$ is projective. Fix $d \geq 1$ and an elementary abelian p -subgroup $H \subseteq \mathfrak{S}_d$. On each H -orbit, the action is regular for a quotient H/K_i , because H is abelian. Choose identifications $H/K_i \simeq E_{a_i}$, allowing $a_i = 0$ for singleton orbits. Regrouping tensor factors identifies $P^{\otimes d}$ with the tensor product of the corresponding orbit powers. Each evaluated orbit power is projective for its quotient group, by the assumption when $a_i > 0$, and trivially when $a_i = 0$. Lemma 5.2 applies to the independent product of the quotient groups. The homomorphism $H \rightarrow \prod_i H/K_i$ is injective, since the permutation action of H is faithful. Restriction along it shows that $\omega(P^{\otimes d})$ is projective over kH . Chouinard's projectivity theorem now implies that $\omega(P^{\otimes d})$ is projective over $k\mathfrak{S}_d$ [5]; see also [3, p. 20].

The tensor power of $P \twoheadrightarrow \mathbf{1}$ is an $\mathfrak{S}_d$ -equivariant epimorphism to the unit with its trivial action. Exact evaluation, followed by coinvariants, therefore gives an epimorphism

$$\omega(P^{\otimes d})_{\mathfrak{S}_d} \twoheadrightarrow \omega(\mathbf{1}) \neq 0.$$

Lemma 5.1 shows that the total symmetrizer $b_d = \sum_{\sigma \in \mathfrak{S}_d} \sigma$ acts nontrivially on $\omega(P^{\otimes d})$, and hence on $P^{\otimes d}$.

On the other hand, the action maps

$$k\mathfrak{S}_d \longrightarrow \text{End}_{\mathcal{C}}(P^{\otimes d})$$

cannot all be injective. To see this without any dimension theory, write $\ell(X)$ for the length of X and put $b = \max_i \ell(S_i \otimes P)$. Exactness gives $\ell(P^{\otimes d}) \leq \ell(P)b^{d-1}$. Since the simples have endomorphism field k , induction on two composition series gives $\dim_k \text{Hom}(X, Y) \leq \ell(X)\ell(Y)$. Thus $\dim_k \text{End}(P^{\otimes d})$ grows at most exponentially, whereas $\dim_k k\mathfrak{S}_d = d!$.

For clarity, the kernels form a tensor ideal in the permutation category $\mathbf{Sym}$: its objects are the nonnegative integers, its endomorphism algebra at d is $k\mathfrak{S}_d$, other morphism spaces are zero, and tensor product is addition with the usual block inclusions of symmetric groups. The symmetry of $\mathcal{C}$ gives a monoidal functor $\mathbf{Sym} \rightarrow \mathcal{C}$, $d \mapsto P^{\otimes d}$, and the kernels form its nonzero kernel ideal. By [8, Lemma 4.10], every nonzero tensor ideal in $\mathbf{Sym}$ contains some total symmetrizer b_d in positive characteristic. Its proof uses Kleshchev's faithfulness result in the appendix to that paper [8, Appendix A, Proposition A.1]. This contradicts the preceding nonvanishing. The least r therefore exists.

Finally, let $H < E_r$ be proper, of rank $a < r$. Its action on the regular E_r -set is a disjoint union of regular H -orbits. Each orbit power has projective evaluation by minimality when $a > 0$, and trivially when $a = 0$. Apply Lemma 5.2 to the independent orbit groups, then restrict to their diagonal copy of H . This proves the last assertion, including $r = 1$. $\square$

The proposition supplies a nonprojective E_r -module whose restriction to every proper subgroup is projective. The next step uses its support to choose the embedding $E_r \hookrightarrow \mathbb{G}_a$.

6 Selecting an embedding by rank support

The preceding section supplies a nonprojective representation of an elementary abelian group whose restrictions to all proper subgroups are projective. We now choose an embedding of that group in the unipotent subgroup of SL_2 so that its tensor product with the restricted Steinberg module remains nonprojective. This choice will make the higher Frobenius functor exact.

Let k be algebraically closed of characteristic $p > 0$, let $E_r = (\mathbb{Z}/p\mathbb{Z})^r$, and fix generators $g_1, \dots, g_r$. For a finite-dimensional kE_r -module D , define

$$\mathrm{Supp}(D) = \left\{ [\alpha_1 : \dots : \alpha_r] \in \mathbb{P}^{r-1}(k) : D \text{ is not free over } k[t]/(t^p), \quad t \mapsto \sum_{i=1}^r \alpha_i(g_i - 1) \right\}.$$

This is the projectivized rank variety. The rank-variety projectivity and tensor-product theorems state that

$$\mathrm{Supp}(D) = \emptyset \iff D \text{ is projective}, \quad \mathrm{Supp}(D \otimes D') = \mathrm{Supp}(D) \cap \mathrm{Supp}(D'). \quad (6.1)$$

These statements hold with the indicated coordinates for every prime; see [4, Sections 1.9 and 1.12] and [10, Section 2.3].

Lemma 6.1. *Suppose that M is a nonprojective finite-dimensional kE_r -module and that $M|_H$ is projective for every proper subgroup $H < E_r$. Every point $[\alpha_1 : \dots : \alpha_r]$ of $\mathrm{Supp}(M)$ has $\mathbb{F}_p$ -linearly independent coordinates.*

Proof. For a nonzero linear form $c : E_r \rightarrow \mathbb{F}_p$, write $c(g_i) = c_i$ and put $H = \ker c$. The tensor identity gives

$$M \otimes \mathrm{Ind}_H^{E_r} k \cong \mathrm{Ind}_H^{E_r}(M|_H),$$

which is projective. We compute the support of $N = \mathrm{Ind}_H^{E_r} k$ explicitly. This is the regular representation of E_r/H , identified with $k[z]/(z^p)$, where a generator of the quotient acts by $1 + z$. Thus $\sum_i \alpha_i(g_i - 1)$ acts by multiplication by

$$h_\alpha(z) = \sum_i \alpha_i((1 + z)^{c_i} - 1),$$

with c_i represented by an integer in $\{0, \dots, p-1\}$. Its linear coefficient is $\sum_i c_i \alpha_i$. If this coefficient is nonzero, substitution $t \mapsto h_\alpha(z)$ is an isomorphism of truncated polynomial algebras, so N is free over $k[t]/(t^p)$. If it is zero, then $h_\alpha \in (z^2)$ and $h_\alpha^{p-1} = 0$, which is impossible on a free module of dimension p . Consequently

$$\text{Supp}(N) = \{[\alpha] : \sum_i c_i \alpha_i = 0\}.$$

Equation (6.1) shows that this hyperplane misses $\text{Supp}(M)$. Varying the nonzero vector $(c_1, \dots, c_r) \in \mathbb{F}_p^r$ proves the assertion. $\square$

Every support point of the minimal-rank module therefore has $\mathbb{F}_p$ -linearly independent coordinates. To make one of them survive after tensoring with a restricted Steinberg module, we next compute the Steinberg support in terms of the embedding parameters.

Lemma 6.2. *Let $\lambda_1, \dots, \lambda_r \in k$ be linearly independent over $\mathbb{F}_p$, and restrict SL_2 -representations along*

$$\iota_\lambda : E_r \longrightarrow \text{SL}_2(k), \quad g_i \longmapsto \begin{pmatrix} 1 & \lambda_i \\ 0 & 1 \end{pmatrix}.$$

Write R_λ for this restriction. Then

$$\text{Supp}(R_\lambda(\text{St}_{r-1})) = \left\{ [\alpha] : \sum_i \alpha_i \lambda_i^{p^j} = 0 \quad (0 \leq j \leq r-2) \right\}. \quad (6.2)$$

Proof. This is the Steinberg case of [10, Proposition 4.2.5 and Corollary 4.2.7]. Here is a direct calculation in the chosen coordinates. On $\text{St}_1 = \text{Sym}^{p-1}(k^2)$, the upper unipotent element with parameter a acts by $\exp(aN) = \sum_{b=0}^{p-1} a^b N^b / b!$, where N has one Jordan block of length p . Therefore $\sum_i \alpha_i (g_i - 1)$ acts by a polynomial in N whose linear coefficient is $\sum_i \alpha_i \lambda_i$. As in the proof of Lemma 6.1, the resulting module of dimension p is free exactly when this coefficient is nonzero. On the j -th Frobenius twist, the same calculation replaces λ_i by $\lambda_i^{p^j}$. Now use

$$\text{St}_{r-1} \cong \bigotimes_{j=0}^{r-2} \text{St}_1^{(j)}$$

and the tensor-product formula in (6.1). When $r = 1$, $\text{St}_0 = k$ has support $\mathbb{P}^0(k)$, agreeing with the empty system in (6.2). $\square$

Thus a point $[\alpha]$ lies in the Steinberg support precisely when the embedding parameters satisfy the equations in (6.2). The next lemma solves these equations while keeping the parameters linearly independent over $\mathbb{F}_p$, as required for an embedding.

Lemma 6.3. *Let $\alpha_1, \dots, \alpha_r \in k$ be linearly independent over $\mathbb{F}_p$. There are $\mathbb{F}_p$ -linearly independent $\lambda_1, \dots, \lambda_r \in k$ such that*

$$\sum_{i=1}^r \alpha_i \lambda_i^{p^j} = 0 \quad (0 \leq j \leq r-2). \quad (6.3)$$

For $r = 1$ the list of equations is empty.

Proof. When $r = 1$, take any $\lambda_1 \neq 0$. Suppose $r \geq 2$. Because k is perfect, we may put $\beta_i = \alpha_i^{p^{-(r-2)}}$. Form the $(r-1) \times r$ matrix

$$B = (\beta_i^{p^s})_{\substack{0 \leq s \leq r-2 \\ 1 \leq i \leq r}}.$$

Its rows are linearly independent. Indeed, a nontrivial row relation would give a nonzero additive polynomial $f(X) = \sum_{s=0}^{r-2} a_s X^{p^s}$ vanishing on the $\mathbb{F}_p$ -span of the β_i . That span has p^r elements, whereas $\deg f \leq p^{r-2}$.

The row space of B contains no nonzero vector in $\mathbb{F}_p^r$. To see this, suppose $c_i = f(\beta_i) \in \mathbb{F}_p$ for such a polynomial f . The additive polynomial $f^p - f$ then vanishes on the same p^r elements and has degree at most p^{r-1} . Thus $f^p - f = 0$, forcing $f = 0$ because f has zero constant term.

Choose a nonzero vector $\lambda \in \ker B$. This kernel is a line. If $c \in \mathbb{F}_p^r$ satisfied $\sum_i c_i \lambda_i = 0$, it would belong to the annihilator of $\ker B$, which is the row space of B . Hence $c = 0$, proving the required independence of the λ_i . The rows of B , in reverse order, are $(\alpha_i^{p^{-j}})_i$ for $0 \leq j \leq r-2$. Raising their nullvector equations to the p^j -th power gives (6.3). $\square$

Proposition 6.4. *Let M be a nonprojective finite-dimensional kE_r -module whose restriction to every proper subgroup is projective. There is an embedding $\iota_\lambda : E_r \hookrightarrow \mathrm{SL}_2(k)$ into the upper unipotent subgroup such that*

$$R_\lambda(\mathrm{St}_{r-1}) \otimes M \quad \text{is nonprojective.}$$

Proof. By (6.1), choose $[\alpha] \in \mathrm{Supp}(M)$. Lemma 6.1 makes its coordinates $\mathbb{F}_p$ -linearly independent. Choose λ by Lemma 6.3. Its independent coordinates make ι_λ injective, and Lemma 6.2 places $[\alpha]$ in the support of $R_\lambda(\mathrm{St}_{r-1})$. The tensor-product theorem now gives the conclusion. $\square$

Theorem 6.5 (An exact higher Frobenius functor). *Let $\mathcal{C}$ be a finite symmetric tensor category over an algebraically closed field k of characteristic $p > 0$. There are an integer $r \geq 1$ and an embedding $E_r \hookrightarrow U(k)$ for which the construction of Section 4 gives an exact additive strong symmetric monoidal functor*

$$\Phi : \mathcal{C} \longrightarrow \mathcal{C} \boxtimes \mathrm{Ver}_{p^r}(k).$$

It is given by $\Phi(X) = \Theta_{\mathcal{C}}(X^{\otimes p^r})$, where Θ is the functor in (4.1) for the chosen embedding. The functor Φ satisfies $\Phi(af) = a^{p^r} \Phi(f)$, and its evaluation at every projective $Q \in \mathcal{C}$ is given by the natural isomorphism (4.4).

Proof. Choose a projective generator P and the least rank r of Proposition 5.3. Then

$$M = \mathrm{Hom}_{\mathcal{C}}(P, P^{\otimes p^r})$$

is nonprojective and has projective restriction to every proper subgroup of E_r . Proposition 6.4 supplies an embedding ι_λ for which $R_\lambda(\mathrm{St}_{r-1}) \otimes M$ remains nonprojective. Theorem 3.1 gives $R_\lambda(\mathrm{St}_{r-1}) \otimes M \cong R_\lambda(T)$ for a tilting module T . Lemma 4.1 therefore gives

$$0 \neq \Theta(R_\lambda(T)) \cong \Theta(R_\lambda(\mathrm{St}_{r-1})) \otimes \Theta(M).$$

In particular $\Theta(M) \neq 0$. Proposition 4.3, with both projectives equal to P , proves that Φ is exact. Its remaining properties and the projective-evaluation identity were established in Section 4. All steps include $p = 2$ and $r = 1$. $\square$

7 Removing the original category from the target

Choose an exact higher Frobenius functor by Theorem 6.5. Write $q = p^r$, $\mathcal{W} = \text{Ver}_q(k)$, and retain

$$\Phi : \mathcal{C} \longrightarrow \mathcal{C} \boxtimes \mathcal{W}.$$

The functor is q -power semilinear, exact, and strong symmetric monoidal. Its evaluation property says that, for every projective $Q \in \mathcal{C}$,

$$(\text{Hom}_{\mathcal{C}}(Q, -) \boxtimes \text{Id})(\Phi(X)) \simeq \Theta(\text{Hom}_{\mathcal{C}}(Q, X^{\otimes q})). \quad (7.1)$$

Here $\Theta : \text{Rep}_k(E_r) \rightarrow \mathcal{W}$ is the k -linear strong symmetric monoidal functor used in its construction. Our objective is to show that sufficiently many iterations of Φ place the first factor in the Serre closure of a Frobenius-exact subcategory. The Serre closure is the smallest full subcategory closed under subobjects, quotients, and extensions that contains it.

7.1 Ordinary Frobenius and a Grothendieck-ring identity

For an object $Y \in \mathcal{C}$, let c denote the cyclic permutation of the p factors of $Y^{\otimes p}$, and set $D = 1 - c$. Thus $D^p = 0$. The components of the ordinary Frobenius functor are

$$F_i(Y) = \frac{\ker D \cap \text{im } D^{i-1}}{\ker D \cap \text{im } D^i}, \quad 1 \leq i < p.$$

If ℓ_i is the simple object of $\text{Ver}_p(k)$ represented by a Jordan block of length i , the functor is

$$F(Y) = \bigoplus_{i=1}^{p-1} F_i(Y) \boxtimes \ell_i.$$

We use the semilinear convention, so the objects $F_i(Y)$ belong to $\mathcal{C}$. This construction and its monoidal structure are due to Etingof–Ostrik [18, Sections 3–5]. It need not be exact.

Let $K_0(\mathcal{C})$ be the Grothendieck ring, with a basis given by simple classes, addition induced by short exact sequences, and multiplication induced by tensor product. Symmetry makes it commutative. Frobenius–Perron dimension d is its unique positive normalized real character: it is additive on exact sequences, multiplicative on tensor products, and equals one on the unit. Each simple object has dimension at least one. On a Deligne product it is the product of the two dimensions; see [17, Sections 3.3 and 4.5]. We need two results from [18, Propositions 6.3 and 7.4]:

$$d(F(Y)) \leq d(Y), \quad \mathcal{C}_{\text{ex}} = \{Y \in \mathcal{C} : d(F(Y)) = d(Y)\}. \quad (7.2)$$

The ordinary Frobenius functor is exact on $\mathcal{C}_{\text{ex}}$, which is a tensor subcategory. Such a category is called *Frobenius exact*. These statements require only finitely many simple isomorphism classes. The tensor subcategory is closed under subquotients and duals; extension closure is not asserted here.

The next identity will connect the two Frobenius constructions: a composition factor of $Y^{\otimes p^r}$ whose multiplicity is nonzero modulo p must occur in an iterated ordinary Frobenius component.

Lemma 7.1. *In the commutative ring $K_0(\mathcal{C})/pK_0(\mathcal{C})$, one has*

$$[Y]^p = \sum_{i=1}^{p-1} i[F_i(Y)].$$

Consequently, for every $r \geq 1$,

$$[Y]^{p^r} = \sum_{i_1, \dots, i_r=1}^{p-1} (i_1 \cdots i_r) [F_{i_r} \cdots F_{i_1}(Y)] \pmod{p}. \quad (7.3)$$

Proof. Put $Z = Y^{\otimes p}$ and $K_j = \ker D \cap \operatorname{im} D^j$, for $0 \leq j \leq p$. The exact sequences

$$0 \longrightarrow K_j \longrightarrow \operatorname{im} D^j \xrightarrow{D} \operatorname{im} D^{j+1} \longrightarrow 0 \quad (0 \leq j < p)$$

give $[Z] = \sum_{j=0}^{p-1} [K_j]$. Since $K_0 \supseteq \cdots \supseteq K_p = 0$, summing the successive quotients in these filtrations gives the integral identity

$$[Z] = \sum_{i=1}^{p-1} i[F_i(Y)] + p[K_{p-1}].$$

Reduction modulo p proves the first assertion. For the second, raise this identity to successive p -th powers in the commutative ring $K_0(\mathcal{C})/p$, and apply the first assertion to each concrete object $F_{i_s} \cdots F_{i_1}(Y)$. This argument does not apply a nonexact functor to a Grothendieck-group relation. $\square$

7.2 Equality in dimension forces Frobenius exactness

An exact strong monoidal functor between finite tensor categories preserves Frobenius–Perron dimension, also when its semilinearity is a field automorphism. Indeed, its map on Grothendieck rings preserves the unit, products, and positive classes. It sends nonzero objects to nonzero objects: apply exactness to the monomorphism $\mathbf{1} \rightarrow X \otimes X^*$ for $X \neq 0$. Composing with dimension therefore gives a positive normalized character; the Perron–Frobenius characterization makes it the dimension character of the source [17, Propositions 3.3.6, 3.3.13, and 4.5.7]. In particular,

$$d(\Phi(X)) = d(X).$$

The simple objects of $\mathcal{C} \boxtimes \mathcal{W}$ are precisely $A \boxtimes B$, with A, B simple, and their dimensions are $d(A)d(B)$.

Lemma 7.2. *Let S be a simple object of $\mathcal{C}$, and let $A \boxtimes B$ be a composition factor of $\Phi(S)$. Then $d(A) \leq d(S)$. If equality holds, then*

$$\Phi(S) \simeq A \boxtimes B, \quad B \text{ is invertible}, \quad S \in \mathcal{C}_{\text{ex}}.$$

In this case A is the unique first component of a composition factor of $\Phi(S)$.

Proof. Every nonzero object has dimension at least one. Thus $d(S) = d(\Phi(S)) \geq d(A)d(B) \geq d(A)$. If the two ends are equal, positivity of the dimensions of all composition factors shows that $\Phi(S)$ has length one and $d(B) = 1$. A simple object of dimension one is invertible, since $B \otimes B^*$ has dimension one and contains the unit.

Let Q_A be the projective cover of A . Over the algebraically closed field k ,

$$\dim_k \operatorname{Hom}_{\mathcal{C}}(Q_A, X) = [X : A]$$

for every X : the functor is exact and has dimension one on A and zero on every other simple object. Applying (7.1) to S gives

$$\Theta(\mathrm{Hom}_{\mathcal{C}}(Q_A, S^{\otimes q})) \simeq B.$$

A symmetric monoidal functor preserves categorical dimension. This dimension of the representation on the left is its vector-space dimension viewed in k , whereas the categorical dimension of the invertible object B is nonzero: its product with the dimension of B^* is one. Consequently,

$$[S^{\otimes q} : A] \not\equiv 0 \pmod{p}. \quad (7.4)$$

This uses categorical dimension only to establish the congruence; the real-valued dimension d remains Frobenius–Perron dimension.

By (7.3) and (7.4), there are indices $i_1, \dots, i_r \in \{1, \dots, p-1\}$ such that A is a composition factor of Y_r , where $Y_0 = S$ and $Y_t = F_{i_t}(Y_{t-1})$. For every t , Equation (7.2) and $d(\ell_{i_t}) \geq 1$ give

$$d(Y_{t-1}) \geq d(F(Y_{t-1})) \geq d(Y_t).$$

Since $d(Y_r) \geq d(A) = d(S)$, all inequalities are equalities. In particular $d(F(S)) = d(S)$, so $S \in \mathcal{C}_{\mathrm{ex}}$. $\square$

The finite graph. We adapt the finite-simple dimension descent used by Etingof–Ostrik for ordinary Frobenius iteration [18, Proposition 10.1] to the exact functor Φ . Let N be the number of simple isomorphism classes in $\mathcal{C}$. Form a directed graph on these classes, putting an arrow $S \rightarrow A$ when $A \boxtimes B$ occurs in $\Phi(S)$ for some simple B . Every vertex has an outgoing arrow. By Lemma 7.2, dimension is nonincreasing along arrows. On a directed cycle it is constant, every vertex belongs to $\mathcal{C}_{\mathrm{ex}}$, and every vertex has just one outgoing neighbor. Thus a path entering a cycle cannot leave it. A path of N edges visits $N+1$ vertices, so it contains such a cycle. Its endpoint therefore belongs to $\mathcal{C}_{\mathrm{ex}}$.

Figure 1 illustrates this trapping mechanism: the unique outgoing neighbor at each cycle vertex prevents a path from leaving the cycle, even though dimension need not decrease strictly before the cycle.

The graph controls the possible first components under repeated application of Φ . To turn its path bound into a statement about an actual tensor functor, we must iterate Φ with compatible scalar actions on all the factors.

7.3 Scalar twists and iteration

Put $\sigma(a) = a^q$ for $a \in k$, where $q = p^r$ is the power chosen above. Since k is algebraically closed, σ is an automorphism. A functor is σ -semilinear if $F(af) = \sigma(a)F(f)$ for every scalar a and morphism f . We now define the iteration of Φ without mixing different scalar actions.

For a k -linear category $\mathcal{A}$, let $\mathcal{A}^{[\sigma]}$ have the same objects and additive morphism groups, with scalar action

$$a \cdot_{[\sigma]} f = \sigma^{-1}(a)f.$$

Thus a σ -semilinear functor $\mathcal{A} \rightarrow \mathcal{B}$ is precisely a k -linear functor $\mathcal{A}^{[\sigma]} \rightarrow \mathcal{B}$. For finite abelian categories the universal property of the Deligne product gives

$$(\mathcal{A} \boxtimes \mathcal{B})^{[\sigma]} \simeq \mathcal{A}^{[\sigma]} \boxtimes \mathcal{B}^{[\sigma]}.$$

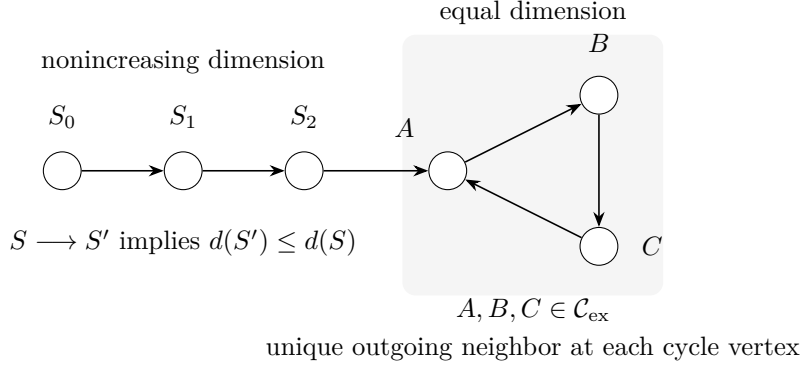


Figure 1: A schematic path in the graph of simple objects, with an illustrative three-vertex cycle. Dimension is nonincreasing along every arrow; arrows before the cycle may also preserve dimension. On a cycle, Lemma 7.2 gives equality, membership in $\mathcal{C}_{\text{ex}}$, and a unique outgoing neighbor at every vertex, so the path cannot leave the cycle.

Consequently two functors with the same semilinearity have a well-defined Deligne product with that semilinearity: apply the ordinary Deligne product to their k -linear versions after this identification.

Every higher Verlinde category is defined over the prime field [2, Section 4.9]; the resulting scalar equivalences are also recorded in [11, Remark 2.3.8]. Hence, for every integer a , scalar transport along $b \mapsto b^{p^a}$ gives an exact strong symmetric semilinear autoequivalence of $\text{Ver}_{p^s}(k)$. Here negative powers denote inverse Frobenius automorphisms of k . Indeed, scalar transport acts on a presentation of the category and its tensor structure over $\mathbb{F}_p$; all structure maps are preserved because the automorphism fixes $\mathbb{F}_p$. Its inverse is transport along the inverse field automorphism. These autoequivalences are scalar transports, distinct from the ordinary Frobenius functor, which need not be exact.

Let J_σ denote such an autoequivalence of $\mathcal{W} = \text{Ver}_q(k)$ with semilinearity σ . Set $\Psi_0 = \text{Id}_{\mathcal{C}}$ and $\Psi_1 = \Phi$. For $j \geq 1$, define

$$\Psi_{j+1} = (\Phi \boxtimes J_\sigma^{\boxtimes j}) \circ \Psi_j : \mathcal{C} \longrightarrow \mathcal{C} \boxtimes \mathcal{W}^{\boxtimes(j+1)}. \quad (7.5)$$

Each factor in the parenthesized functor has semilinearity σ , so the product is defined as above. By induction Ψ_j is exact, strong symmetric monoidal, and σ^j -semilinear. The old $\mathcal{W}$ -factors are only transported by equivalences. We now apply the graph argument to the first components of its composition factors.

7.4 The image after iteration

The scalar-compatible iteration (7.5) gives an exact functor

$$\Psi_N : \mathcal{C} \longrightarrow \mathcal{C} \boxtimes \mathcal{W}^{\boxtimes N}.$$

Exactness is essential here: applying the next functor to a composition series shows that every first component of every composition factor follows a path of one additional edge. The autoequivalences on the other factors send simple objects to simple objects and do not change the first component. Induction, first for simple inputs and then for a composition series of an arbitrary input, shows that all first components of $\Psi_N(X)$ belong to $\mathcal{C}_{\text{ex}}$.

Let $\mathcal{D}$ be the full subcategory of $\mathcal{C}$ consisting of objects whose simple composition factors belong to $\mathcal{C}_{\text{ex}}$. This is the Serre closure of $\mathcal{C}_{\text{ex}}$. It is closed under tensor products, because biexactness gives tensor-product filtrations and $\mathcal{C}_{\text{ex}}$ is a tensor subcategory closed under subquotients. Exact duality also preserves it. Hence $\mathcal{D}$ inherits the rigid symmetric tensor structure and the simple unit.

For completeness, the relevant abelian and Deligne-product assertions have a concrete finite-algebra description. Choose a basic finite dimensional algebra H with $\mathcal{C} \simeq H\text{-mod}$, and let e be the sum of the primitive idempotents corresponding to the simple objects outside $\mathcal{C}_{\text{ex}}$. The functor $M \mapsto eM$ is exact, and vanishes on exactly the allowed simples. Thus

$$\mathcal{D} \simeq (H/HeH)\text{-mod}.$$

In particular $\mathcal{D}$ is finite and has enough projectives. If $\mathcal{W}^{\boxtimes N} \simeq J\text{-mod}$, then

$$\mathcal{D} \boxtimes \mathcal{W}^{\boxtimes N} \simeq ((H/HeH) \otimes_k J)\text{-mod}$$

is the full Serre subcategory of $(H \otimes_k J)\text{-mod}$ annihilated by $e \otimes 1$. It consists exactly of objects with allowed first components. We have therefore obtained an exact strong symmetric monoidal functor

$$\Psi_N : \mathcal{C} \longrightarrow \mathcal{D} \boxtimes \mathcal{W}^{\boxtimes N}. \quad (7.6)$$

The remaining step is to fiber $\mathcal{D}$ over a small Verlinde category and to remove the total scalar twist.

7.5 A fiber functor on the Serre closure

It remains to replace $\mathcal{D}$ in Equation (7.6) by a fixed low Verlinde level and correct the scalar action.

We first check finiteness for the subcategories in the fiber-functor theorem. More generally, let $\mathcal{B}$ be a full additive subcategory of $A\text{-mod}$ closed under subobjects and quotients, where A is finite dimensional. Put $I = \bigcap_{M \in \mathcal{B}} \text{Ann}_A(M)$. Since A is finite dimensional, this intersection is already a finite intersection. A finite direct sum $M \in \mathcal{B}$ therefore satisfies $\text{Ann}_A(M) = I$. Give $\text{End}_k(M)$ the left A -action by postcomposition: $a \cdot u = \rho_M(a) \circ u$. The action map embeds the left regular A/I -module in $\text{End}_k(M) \cong M^{\oplus \dim_k M}$, so $A/I \in \mathcal{B}$. Every finite A/I -module is a quotient of a finite direct sum of A/I . Thus $\mathcal{B} = (A/I)\text{-mod}$ and, in particular, $\mathcal{B}$ is finite abelian and has enough projectives. This applies to $\mathcal{C}_{\text{ex}}$, which is a tensor subcategory closed under subquotients [18, Proposition 7.4]. The inherited tensor product and duality therefore make $\mathcal{C}_{\text{ex}}$ a finite symmetric tensor category; these properties for $\mathcal{D}$ were established above.

By [18, Theorem 8.1], the Frobenius-exact category $\mathcal{C}_{\text{ex}}$ has a k -linear fiber functor to $\text{Ver}_p(k)$. If $p > 2$, its Serre closure equals $\mathcal{C}_{\text{ex}}$ [18, Corollary 9.5]; hence the same is true of $\mathcal{D}$.

If $p = 2$, the preceding fiber functor has target $\text{Ver}_2(k) = \text{Vec}_k$. The finite tensor subcategory $\mathcal{C}_{\text{ex}} \subseteq \mathcal{D}$ contains every simple object of $\mathcal{D}$, by the definition of $\mathcal{D}$. Thus the extension theorem of Etingof–Gelaki [16, Theorem 2.21(2)] applies to this inclusion and gives a k -linear fiber functor $\mathcal{D} \rightarrow \mathcal{V}$, exactly as in [18, Corollary 9.8]. Here

$$\mathcal{V} = \text{Rep}_k(k[d]/(d^2), \Delta(d) = d \otimes 1 + 1 \otimes d, R = 1 \otimes 1 + d \otimes d).$$

In that result, “almost Frobenius exact” means precisely the existence of this exact faithful symmetric tensor functor [18, Definition 8.13]. The category $\mathcal{V}$ is the category $\mathcal{C}_1$ of [1, Introduction]; by [2, Theorem 4.5(iii)],

$$\mathcal{V} \simeq \text{Ver}_4^+(k) \subset \text{Ver}_4(k)$$

as symmetric tensor categories. This is the only additional level needed in characteristic two.

7.6 Completion of the fiber functor

Set $n = \max(2, r)$ and $\mathcal{Z} = \text{Ver}_{p^n}(k)$. The exact symmetric tower inclusions [2, Theorem 4.34], together with the preceding fiber functor, give k -linear exact strong symmetric functors

$$E : \mathcal{D} \longrightarrow \mathcal{Z}, \quad I : \mathcal{W} \longrightarrow \mathcal{Z}.$$

They combine to a k -linear exact strong symmetric functor

$$H : \mathcal{D} \boxtimes \mathcal{W}^{\boxtimes N} \longrightarrow \mathcal{Z}, \quad H(D \boxtimes W_1 \boxtimes \cdots \boxtimes W_N) = E(D) \otimes I(W_1) \otimes \cdots \otimes I(W_N). \quad (7.7)$$

Exactness and the symmetric tensor structure follow from Lemma 2.2. The composite $H\Psi_N$ has semilinearity σ^N . Let $J_{\sigma^{-N}}$ be scalar transport on $\mathcal{Z}$ with semilinearity σ^{-N} . Then

$$\mathcal{F} = J_{\sigma^{-N}} \circ H \circ \Psi_N : \mathcal{C} \longrightarrow \text{Ver}_{p^n}(k)$$

is k -linear, exact, and strong symmetric monoidal. Finally, an exact strong monoidal functor between tensor categories with simple units is faithful. Indeed, if $X \neq 0$, the coevaluation $\mathbf{1} \rightarrow X \otimes X^*$ is nonzero by a triangle identity, hence injective by simplicity of $\mathbf{1}$. Its exact image shows that $\mathcal{F}(X) \neq 0$. For a nonzero morphism f , exactness gives $\text{im } \mathcal{F}(f) \cong \mathcal{F}(\text{im } f) \neq 0$. Thus $\mathcal{F}$ is the required fiber functor, proving Theorem 1.1.

8 Classification, uniqueness, and reconstruction

We combine Theorem 1.1 with the established structure of the higher Verlinde categories and generalized Tannaka duality. The resulting statements concern finite symmetric tensor categories; the uniqueness assertions apply in characteristic two.

8.1 Finite incompressible symmetric tensor categories

For a finite symmetric tensor category $\mathcal{C}$ over k , *incompressibility* means that every k -linear exact strong symmetric monoidal functor from $\mathcal{C}$ to a symmetric tensor category over k is fully faithful. This is the finite-source form of incompressibility in [9, Section 3.2]. The symmetry condition is part of the definition here.

Write Vec_k for finite-dimensional vector spaces and, when $p > 2$, sVec_k for finite-dimensional supervector spaces with the sign symmetry. For $p^m > 2$, $\text{Ver}_{p^m}^+(k)$ denotes the neutral component of the $\mathbb{Z}/2$ -grading constructed from the even highest-weight tilting modules in [2, Section 4.1]. The low levels satisfy

$$\text{Ver}_2(k) \simeq \text{Vec}_k, \quad \text{Ver}_3(k) \simeq \text{sVec}_k, \quad \text{Ver}_3^+(k) \simeq \text{Vec}_k;$$

see [2, Theorem 4.5(iii), Corollary 4.11, and Example 4.52]. In particular, the plus notation is used only when $p^m > 2$.

Corollary 8.1 (Finite incompressible classification). *Let k be algebraically closed of characteristic $p > 0$. A finite symmetric tensor category over k is incompressible if and only if it is equivalent as a k -linear symmetric tensor category to a category in the corresponding row:*

$$\begin{array}{l|l} p = 2 & \text{Ver}_{2^m}(k) \ (m \geq 1), \quad \text{Ver}_{2^m}^+(k) \ (m \geq 2), \\ p = 3 & \text{Ver}_{3^m}(k), \ \text{Ver}_{3^m}^+(k) \ (m \geq 1), \\ p \geq 5 & \text{Vec}_k, \ \text{sVec}_k, \quad \text{Ver}_{p^m}(k), \ \text{Ver}_{p^m}^+(k) \ (m \geq 1). \end{array}$$

Proof. Let $\mathcal{C}$ be incompressible and choose a fiber functor $F : \mathcal{C} \rightarrow \text{Ver}_{p^n}(k)$ by Theorem 1.1. Let $\mathcal{I}$ be the full subcategory of subquotients of objects $F(X)$, $X \in \mathcal{C}$. Because $\mathcal{C}$ is finite, $\mathcal{I}$ is a finite tensor subcategory and the corestriction $F : \mathcal{C} \rightarrow \mathcal{I}$ is surjective in the tensor-category sense [17, Definition 1.8.18, Proposition 1.8.19, and Section 6.3]. Incompressibility makes this corestriction fully faithful. The injective and surjective Frobenius–Perron-dimension inequalities for finite tensor categories give, respectively,

$$\text{FPdim}(\mathcal{C}) \leq \text{FPdim}(\mathcal{I}), \quad \text{FPdim}(\mathcal{C}) \geq \text{FPdim}(\mathcal{I}).$$

Equality in either inequality makes the corestriction an equivalence [17, Propositions 6.3.3 and 6.3.4].

Corollary 4.61 of [2] now restricts $\mathcal{I}$ to the full and plus higher Verlinde subcategories at heights $m \leq n$. For odd p , the bottom cases are included using the explicit convention $\text{Ver}_1 = \text{sVec}_k$ and $\text{Ver}_1^+ = \text{Vec}_k$ of [9, Section 1.3]. With the low-level identifications above, these are exactly the displayed possibilities.

Conversely, the full and plus categories in the list are incompressible for symmetric tensor functors by [2, Theorem 4.71 and Remark 4.72(iii)]. For Vec_k , fullness follows from $\text{End}(\mathbf{1}) = k$. For sVec_k in odd characteristic, a symmetric tensor functor sends the odd invertible generator to an odd invertible object. It cannot be isomorphic to the unit, since their self-symmetries are respectively -1 and 1 . The functor is therefore fully faithful on the two simple objects, and hence on all their finite direct sums. $\square$

Every finite tensor category has moderate growth: multiplication by the class of a fixed object on the finite-rank Grothendieck group is a finite nonnegative matrix, which bounds the lengths of its tensor powers exponentially. Thus the corollary is the finite case of [9, Conjecture B].

8.2 Uniqueness in characteristic two

For $p = 2$, set

$$\text{Ver}_{2^\infty}(k) := \bigcup_{n \geq 1} \text{Ver}_{2^n}(k),$$

using the exact symmetric tower embeddings. A fiber functor to this union means a k -linear exact faithful strong symmetric monoidal functor, just as for a finite target.

Corollary 8.2 (Uniqueness into the characteristic-two union). *Let k be algebraically closed of characteristic two and let $\mathcal{C}$ be a finite symmetric tensor category over k . There is exactly one fiber functor*

$$\mathcal{C} \longrightarrow \text{Ver}_{2^\infty}(k)$$

up to monoidal natural isomorphism. Its isomorphism class contains a representative that factors through $\text{Ver}_{2^n}(k)$ for some finite $n \geq 1$. For each fixed $n \geq 1$, there is at most one fiber functor $\mathcal{C} \rightarrow \text{Ver}_{2^n}(k)$ up to monoidal natural isomorphism.

Proof. Theorem 1.1 supplies a fiber functor to a finite stage, and composition with the tower inclusion supplies a fiber functor to the union. By [9, Theorem 9.4.1], an arbitrary tensor category in characteristic two has at most one k -linear exact symmetric tensor functor to $\text{Ver}_{2^\infty}(k)$ up to monoidal natural isomorphism. Applying this to $\mathcal{C}$ proves uniqueness in the union. For two fiber functors to the same finite stage, this isomorphism lifts through the fully faithful tower inclusion; its naturality and tensor compatibility hold at that stage by faithfulness. This proves the fixed-stage assertion as well. $\square$

8.3 Generalized Tannaka reconstruction

Let $\mathcal{V}$ be a finite symmetric tensor category over k . An affine group scheme internal to $\mathcal{V}$ is specified by a commutative Hopf algebra $\mathcal{O}(G)$ in $\text{Ind}(\mathcal{V})$. The fundamental group

$$\pi(\mathcal{V}) := \underline{\text{Aut}}^\otimes(\text{Id}_{\mathcal{V}})$$

acts canonically on every object of $\text{Ind}(\mathcal{V})$. Following [9, Sections 4.1.1–4.1.4], a $\mathcal{V}$ -group is an internal affine group scheme G with a homomorphism $\phi : \pi(\mathcal{V}) \rightarrow G$ such that the coaction on $\mathcal{O}(G)$ induced by conjugation through ϕ equals its canonical $\pi(\mathcal{V})$ -coaction. The category $\text{Rep}(G, \phi)$ consists of G -representations in $\mathcal{V}$ whose restriction along ϕ is the canonical $\pi(\mathcal{V})$ -action. Write $\omega : \text{Rep}(G, \phi) \rightarrow \mathcal{V}$ for the forgetful functor.

Corollary 8.3 (Reconstruction over a finite higher Verlinde level). *Let k be algebraically closed of characteristic $p > 0$, and let $\mathcal{C}$ be a finite symmetric tensor category over k . Choose $n \geq 1$ and a fiber functor $F : \mathcal{C} \rightarrow \mathcal{V} = \text{Ver}_{p^n}(k)$ as in Theorem 1.1. There is a $\mathcal{V}$ -group (G, ϕ) and a k -linear symmetric tensor equivalence*

$$E : \mathcal{C} \xrightarrow{\sim} \text{Rep}(G, \phi)$$

with a monoidal natural isomorphism $\omega \circ E \cong F$. One may take

$$G = \underline{\text{Aut}}^\otimes(F), \quad \phi : \pi(\mathcal{V}) \longrightarrow \underline{\text{Aut}}^\otimes(F)$$

to be the canonical homomorphism induced by F .

Proof. The finite-category conventions, including the simple unit supplied by Lemma 2.1, make $\mathcal{C}$ and $\mathcal{V}$ pretannakian categories in the terminology of [9]. The chosen F is a tensor functor in that paper’s sense. Its Theorem 4.2.1 assigns to F the $\mathcal{V}$ -group displayed above, and the inverse assignment is the forgetful functor from $\text{Rep}(G, \phi)$. The resulting equivalence over $\mathcal{V}$ is precisely the one stated here. $\square$

The classification and existence assertions here remain confined to finite categories. Uniqueness is asserted in characteristic two, both in the tower union and, whenever a fiber functor exists, at each fixed finite stage. The existence theorem supplies some finite height depending on $\mathcal{C}$; it does not supply existence at every prescribed height. In the reconstruction, G is asserted to be affine; no finiteness or finite-type property of this group scheme is used or claimed.

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